Pothole Detection Based on Drone using ESP32

Aninnovative approach to automated road infrastructure monitoring

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Introduction & Problem Statement

The Challenge

Road potholes represent a critical infrastructure challenge, causing thousands of vehicle accidents and millions in damages annually. Traditional manual inspection methods are time-consuming,labor-intensive, and often fail to identify hazards before they cause incidents.

Project Objectives



Drone-Based System

Develop a complete drone platform specifically engineered for aerial road monitoring and pothole detection applications



Wireless Control

Implement seamless wireless drone navigation using joystick input processed through ESP32 microcontroller technology



Live Interface

Displayreal-timejoystick telemetry data through an accessible web-based monitoring interface for operator feedback



Enhanced Safety

Improveroadinspection efficiency while eliminating worker exposure to traffic hazards during manual surveys

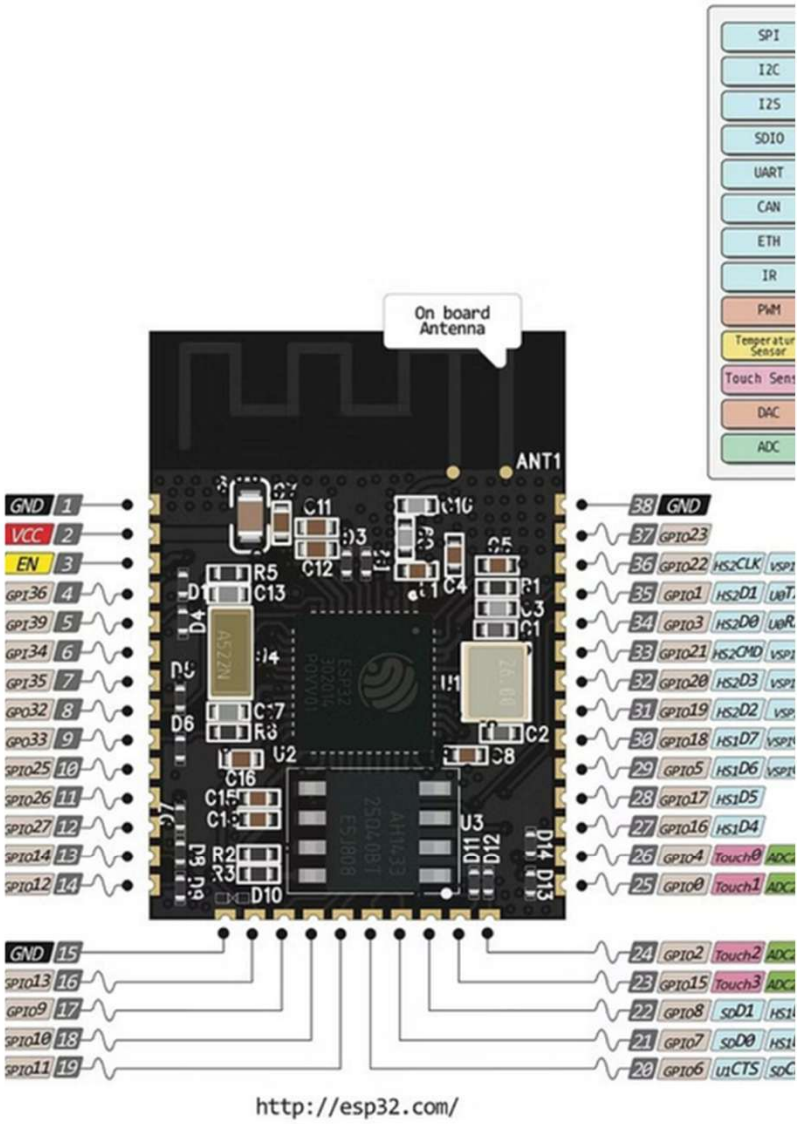
Technical Components

Hardware Components

- ESP32 microcontroller (WiFi/Bluetooth capable)
- Quadcopter frame with brushless motors
- Electronic Speed Controllers (ESCs)
- Analog joystick transmitter module
- Lithium polymer battery pack
- Connecting wires and mounting hardware

Software Stack

- Arduino IDE for ESP32 programming
- Custom WiFi server implementation
- Web interface with HTML /CSS/JavaScript
- ADC signal processing algorithms
- Real-time data visualization



Working Principle

Analog control signals generated by user manipulation of joystick axes

ESP32 Processing
Analog-to-Digital Converter (ADC) reads joystick values with 12-bit resolution

WiFi Transmission
Digital control commands transmitted wirelessly to drone receiver

Drone Response
ESC modules translate signals into motor speed adjustments for flight control

Web Display
Live telemetry values streamed to browser interface for real-time monitoring

Flight Control Parameters

Roll

Controls lateral movement along the drone's longitudinal axis, enabling left and right translation during flight operations

Pitch

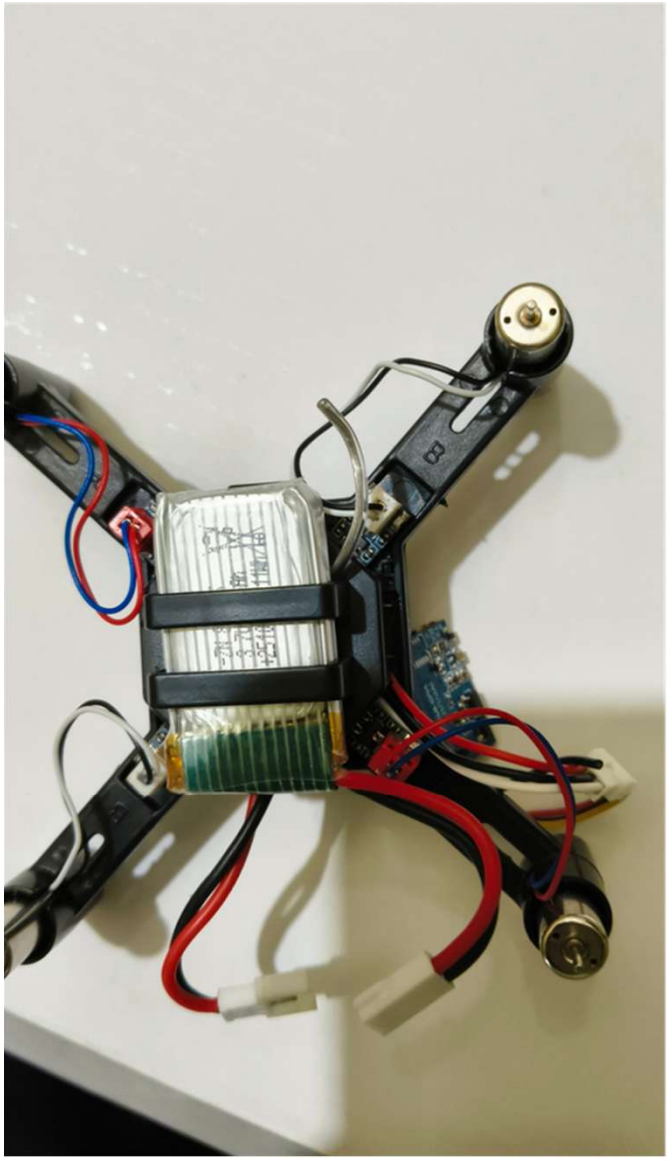
Manages forward and backward motion along the lateral axis, crucial for directional navigation and positioning

Throttle

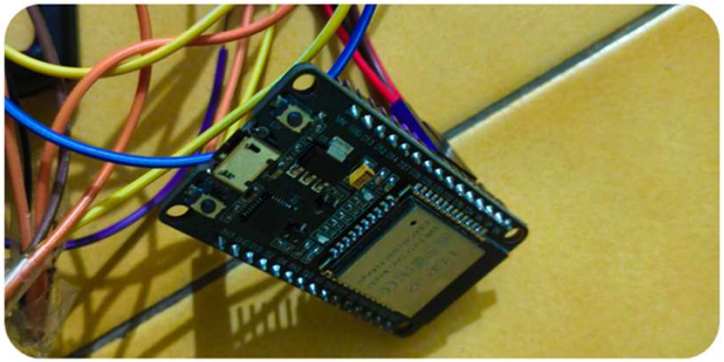
Regulates vertical altitude control by adjusting motor RPM, determining ascent and descent rates for elevation changes

Yaw

Controls rotational movement around the vertical axis, allowing directional changes and orientation adjustments during flight



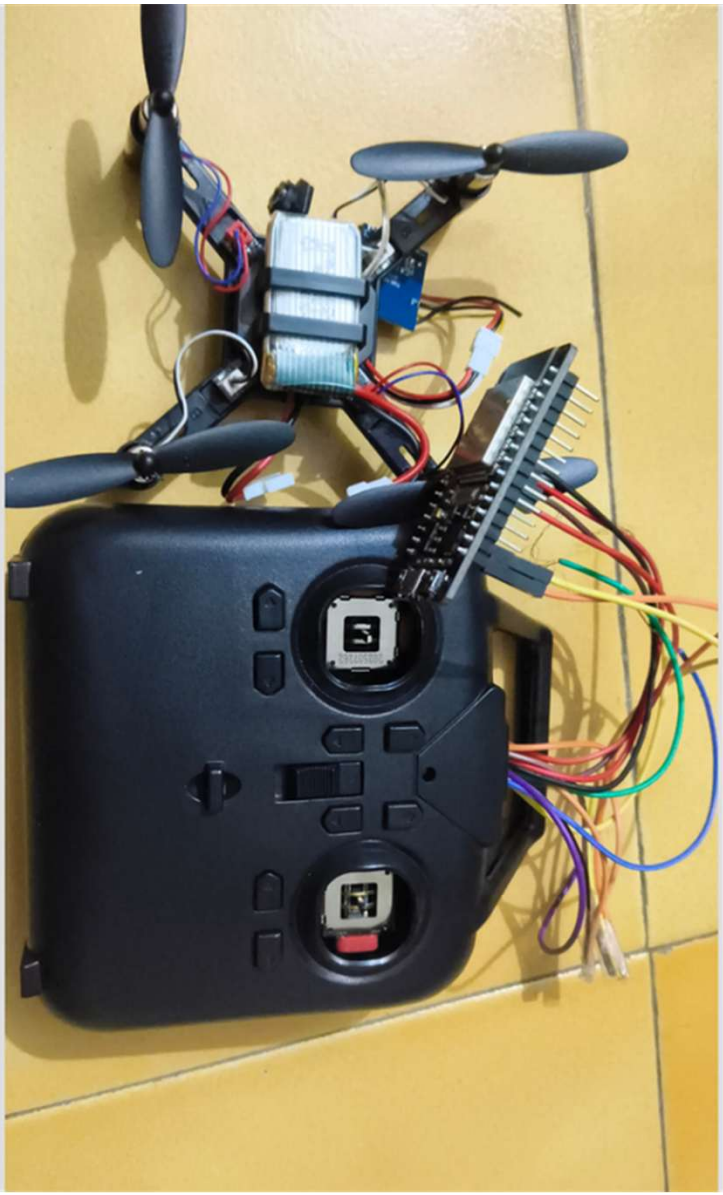
ADC Signal Processing



Analog-to-Digital Conversion

The joystick produces continuousanalog voltage outputs (0-3.3V) that must be converted into discrete digital values for processing. ESP32's 12- bit ADC provides 4,096 quantization levels (0-4095), offering sufficient resolution for precise control.

These digital values are then mapped to PWM signals for ESC control and simultaneously transmitted to the web interface for operator feedback, enabling real-time system monitoring and adjustment.



Result And Future Enhancement

Key Achievment

Successfully established stable WiFi communication

Achieved real-time telemetry display with <50ms latency

Demonstrated smooth joystick response and drone control

Validated system reliability through extended testing

Created accessible web interface for any device

Future Scope

Integrate AI-powered computer vision for automatic pothole detection

Add HD camera module with image processing

Implement GPS waypoint navigation system

Develop autonomous flight modes for routine inspections

Enable cloud-based data storage and analysis

Conclusion

Technical Innovation

The integration of joystick-based analog input with digital processing and web-based visualization represents an accessible approach to infrastructure inspection that reduces costs while improving efficiency

Broader Impact

This system has direct applications for smart city infrastructure management, enabling rapid road condition assessment, preventive maintenance scheduling, and reduced vehicle damage through early hazard detection